General Mathematics — Exercise Sheet 10

L'Hôpital and the shape of a graph

Variant: Statements

Exercise 1 (A point supplied by Cauchy's theorem). Apply Cauchy's mean value theorem to $f(x) = \sin x$ and $g(x) = x^2$ on $[0, \pi/2]$. Show that it supplies a point $c \in (0, \pi/2)$ satisfying $\pi^2 \cos c = 8c$. Prove that this point is unique in that interval, without finding a numerical approximation. Use Lecture 9's monotonicity criterion for the uniqueness step.

Exercise 2 (Choose a justified limit method). Evaluate the following limits. State the limiting form and all division conditions whenever using L'Hôpital. Distinguish the proved zero-over-zero rule from an evaluation using the cited infinity-over-infinity form.

$$\lim_{x\to 0} \frac{\arctan x - x}{x^3}, \qquad \lim_{x\to+\infty} x^2 e^{-x}, \qquad \lim_{x\to 0} (1 + \sin x)^{1/x}.$$

For the last expression, use its positive real base near zero. Lecture 6's standard limits may be used instead of L'Hôpital.

Exercise 3 (A stationary-point family). For $a \in \mathbb{R}$, let $F_a(x) = x^4 - ax^2$ on $\mathbb{R}$. Find all local extrema and inflection points, with their values, for each parameter range. Identify all global minima and decide whether any global maximum exists. Include $a = 0$, where the second-derivative test is inconclusive.

Exercise 4 (A logarithmic graph). Investigate $L(x) = \ln x/x$ on its real domain. Find its axis intersection, all asymptotes, monotonicity intervals, extrema, convexity intervals, inflection points and range. Sketch the graph. You may use the cited limit $\ln x/x \to 0$ at $+\infty$ from Lecture 10, retaining that dependency in the asymptote calculation.

Exercise 5 (The least surface for a closed cylinder). A closed right circular cylinder has fixed volume $V_0 > 0$, radius $r > 0$ and height $h > 0$. Its volume is $\pi r^2 h$ and its total surface area, including both end discs, is $2\pi r^2 + 2\pi rh$. Use the volume constraint to find the unique dimensions minimising surface area. Prove a global minimum on the physical domain, rather than merely finding a stationary point.

